

DISCRETE MATHEMATICS

Introduction To Set Theory

Sets, Operations, Counting and Order

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What is a Set?

Definition

A **set** is a well-defined collection of objects.

The objects belonging to a set are called its **elements** or **members**.

Notation for Sets

Sets are usually denoted by capital letters:

$$A, \quad B, \quad C.$$

Elements are enclosed in braces:

$$A = \{1, 2, 3, 4\}.$$

Thus,

$$1, 2, 3, 4 \in A.$$

Membership

If x is an element of A , we write

$$\boxed{x \in A}.$$

If x is not an element of A , we write

$$\boxed{x \notin A}.$$

Example

If

$$A = \{2, 4, 6, 8\},$$

then

Equality of Sets

Two sets are equal when they contain exactly the same elements.

$$\boxed{A = B}$$

if and only if

$$\boxed{\forall x (x \in A \iff x \in B)}.$$

For example,

$$\{1, 2, 3\} = \{3, 1, 2\}.$$

Order and Repetition

The order of elements does not matter.

$$\{1, 2, 3\} = \{3, 2, 1\}.$$

Repeated elements are also ignored.

$$\{1, 1, 2, 2, 3\} = \{1, 2, 3\}.$$

A set records **membership**, not order or repetition.

Cardinality

The number of elements in a finite set is called its *cardinality*.

It is denoted by

$$|A|.$$

If

$$A = \{2, 4, 6, 8\},$$

then

$$\boxed{|A| = 4}.$$

Set Builder Form

A set can be described by stating a condition that its elements satisfy.

The general form is

$$A = \{x \in U \mid P(x)\}$$

where $P(x)$ is a condition on x .

Reading Set Builder Form

Consider

$$A = \{x \in \mathbb{N} \mid x < 6\}.$$

This means:

The set of all natural numbers x such that $x < 6$.

Example: Natural Numbers

Consider

$$A = \{x \in \mathbb{N} \mid x < 6\}.$$

In roster form,

$$A = \{1, 2, 3, 4, 5\}.$$

The two forms describe the same set.

Example: Even Numbers

The even natural numbers less than 10 can be written as

$$A = \{x \in \mathbb{N} \mid x < 10 \text{ and } 2 \mid x\}.$$

In roster form,

$$A = \{2, 4, 6, 8\}.$$

Types of Sets

Important types of sets include:

- ▶ Empty set
- ▶ Singleton set
- ▶ Finite set
- ▶ Infinite set
- ▶ Subset
- ▶ Proper subset
- ▶ Power set

Empty Set

Definition

The **empty set** is the set containing no elements.

It is denoted by

$$\boxed{\emptyset}.$$

For example,

$$\{x \in \mathbb{N} \mid x < 0\} = \emptyset.$$

Singleton Set

Definition

A **singleton set** contains exactly one element.

For example,

$$A = \{7\}.$$

Therefore,

$$\boxed{|A| = 1}.$$

Finite and Infinite Sets

A set is **finite** if it contains a finite number of elements.

$$A = \{1, 2, 3, 4, 5\}.$$

A set is **infinite** if it is not finite.

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

Subset

Definition

A is a subset of B if every element of A is also an element of B .

We write

$$\boxed{A \subseteq B}.$$

Equivalently,

$$A \subseteq B \iff \forall x (x \in A \Rightarrow x \in B).$$

Subset: Example

Let

$$A = \{1, 2\}$$

and

$$B = \{1, 2, 3, 4\}.$$

Since every element of A belongs to B ,

$$\boxed{A \subseteq B}.$$

Proper Subset

If

$$A \subseteq B$$

and

$$A \neq B,$$

then A is called a proper subset of B .

We write

$$\boxed{A \subset B}.$$

Power Set

Definition

The **power set** of A , denoted by $\mathcal{P}(A)$, is the set of all subsets of A .

For

$$A = \{a, b\},$$

we have

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$$

Cardinality of a Power Set

If

$$|A| = n,$$

then

$$\boxed{|\mathcal{P}(A)| = 2^n.}$$

For example, if

$$|A| = 3,$$

then

Set Operations

The fundamental operations are:

$$A \cup B \quad A \cap B \quad A - B \quad A^c$$

They allow us to construct new sets from existing sets.

Union

Definition

The union of A and B consists of all elements that belong to A or to B .

It is written

$$\boxed{A \cup B}.$$

$$x \in A \cup B \iff (x \in A) \vee (x \in B).$$

Union: Example

Let

$$A = \{1, 2, 3\}$$

and

$$B = \{3, 4, 5\}.$$

Then

$$A \cup B = \{1, 2, 3, 4, 5\}.$$

The common element 3 appears only once.

Intersection

Definition

The intersection of A and B consists of the elements common to both sets.

It is written

$$\boxed{A \cap B}.$$

$$x \in A \cap B \iff (x \in A) \wedge (x \in B).$$

Intersection: Example

Let

$$A = \{1, 2, 3\}$$

and

$$B = \{3, 4, 5\}.$$

Then

$$A \cap B = \{3\}.$$

The intersection contains only the common elements.

Set Difference

Definition

The difference $A - B$ consists of the elements that belong to A but do not belong to B .

$$A - B = \{x \mid x \in A, x \notin B\}.$$

Difference: Example

Let

$$A = \{1, 2, 3, 4\}$$

and

$$B = \{3, 4, 5, 6\}.$$

Then

$$A - B = \{1, 2\}.$$

$$B - A = \{5, 6\}.$$

Complement

Let U be the universal set.

Definition

The complement of A consists of all elements of U that do not belong to A .

It is written

$$A^c.$$

Complement: Example

Let

$$U = \{1, 2, 3, 4, 5, 6\}$$

and

$$A = \{1, 2, 3\}.$$

Then

$$A^c = \{4, 5, 6\}.$$

De Morgan's Laws

Two important identities are

$$(A \cup B)^c = A^c \cap B^c$$

and

$$(A \cap B)^c = A^c \cup B^c.$$

These laws connect complements with union and intersection.

The Counting Problem

Suppose we want to find

$$|A \cup B|.$$

A first attempt might be

$$|A| + |B|.$$

But elements in

$$A \cap B$$

have been counted twice.

The Inclusion–Exclusion Principle

We subtract the overlap once.

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The formula corrects the double counting.

Example

Suppose

$$|A| = 30, \quad |B| = 25, \quad |A \cap B| = 10.$$

Then

$$|A \cup B| = 30 + 25 - 10.$$

Therefore,

$$\boxed{|A \cup B| = 45.}$$

Three Sets

For three sets,

$$A, \quad B, \quad C,$$

we have

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |B \cap C| - |C \cap A| \\ &\quad + |A \cap B \cap C|. \end{aligned}$$

Partially Ordered Sets

A partial order is a relation satisfying:

- ▶ Reflexivity
- ▶ Antisymmetry
- ▶ Transitivity

A set together with a partial order is called a

partially ordered set

or poset.

Hasse Diagrams

A Hasse diagram is a graphical representation of a partially ordered set.

It emphasizes the essential ordering relationships.

Relations that can be inferred through transitivity are not drawn.

Cover Relation

Suppose

$$a < b.$$

If there is no element c such that

$$a < c < b,$$

then b **covers** a .

A line is drawn between a and b .

Simple Hasse Diagram

Consider

$$\{1, 2, 3\}$$

with the usual order.



Hasse Diagram and Divisibility

Consider

$$S = \{1, 2, 3, 6\}$$

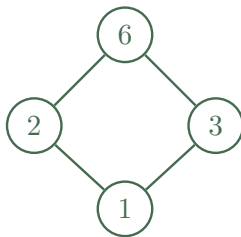
with the relation

$$a \mid b.$$

For example,

$$1 \mid 2, \quad 1 \mid 3, \quad 2 \mid 6, \quad 3 \mid 6.$$

Divisibility Hasse Diagram



Power Set as a Poset

Consider

$$A = \{a, b\}.$$

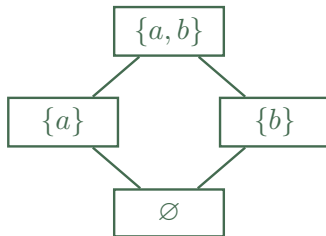
Its power set is

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$$

Order these sets using

$$\subseteq.$$

Hasse Diagram of $\mathcal{P}(\{a, b\})$



The ordering relation is $\subseteq$.

Summary

- ▶ A set is a well-defined collection of objects.
- ▶ Set-builder notation describes sets using conditions.
- ▶ Sets may be empty, finite, infinite, or related by subsets.
- ▶ Union, intersection, difference and complement are fundamental set operations.
- ▶ Inclusion–exclusion provides a method for correcting double counting.
- ▶ Hasse diagrams represent partially ordered sets visually.

Important Formulas

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|\mathcal{P}(A)| = 2^{|A|}$$

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Reference

Primary Reference

Stanford Encyclopedia of Philosophy

Set Theory

<https://plato.stanford.edu/entries/set-theory/>

The presentation uses the reference for the foundational treatment of sets, membership, subsets, set operations and related concepts.

THANK YOU

